Graphs 9 — Gradient of a function · Differentiation · Stationary points · Motion in a straight line

Calculator allowed. Marks in brackets. ★ = harder question. If $y = kx^n$ then $\dfrac{dy}{dx} = nkx^{n-1}$.

1 Find the gradient of the tangent to each curve at $x = 1$. [7]

a $y = 2x^3 + 5x - 1$ **b** $y = (x - 1)(4x + 3)$ **c** $y = 3 - \dfrac{4}{x^2}$

a $\dfrac{dy}{dx} = 6x^2 + 5$; at $x = 1$: 11. **b** Expand first: $y = 4x^2 - x - 3$, $\dfrac{dy}{dx} = 8x - 1$; at $x = 1$: 7. **c** $y = 3 - 4x^{-2}$, $\dfrac{dy}{dx} = 8x^{-3} = \dfrac{8}{x^3}$; at $x = 1$: 8.

2 Find the stationary points of the curve $y = x^3 - 6x^2 + 9x + 1$ and say whether each is a maximum or a minimum. [6]

$\dfrac{dy}{dx} = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3) = 0 \Rightarrow x = 1$ or $x = 3$.

$x = 1$: $y = 1 - 6 + 9 + 1 = 5$. $x = 3$: $y = 27 - 54 + 27 + 1 = 1$. Stationary points $(1, 5)$ and $(3, 1)$.

Classify by the gradient either side: at $x = 0$ the gradient is $9 > 0$, at $x = 2$ it is $-3 < 0$, at $x = 4$ it is $9 > 0$. So $(1, 5)$ is a **maximum** (gradient goes $+\,0\,-$) and $(3, 1)$ a **minimum** ($-\,0\,+$) — the usual shape of a positive cubic.

3 The fuel consumption y km per litre of a car travelling at x km/h is modelled by $y = 60 - \dfrac{x}{4} - \dfrac{1600}{x}$ for $40 \le x \le 120$. **a** Find an expression for $\dfrac{dy}{dx}$. **b** Find the most economical speed and the fuel consumption at that speed. [6]

a $y = 60 - \frac{1}{4}x - 1600x^{-1}$, so $\dfrac{dy}{dx} = -\dfrac{1}{4} + 1600x^{-2} = -\dfrac{1}{4} + \dfrac{1600}{x^2}$.

b Maximum where $\dfrac{dy}{dx} = 0$: $\dfrac{1600}{x^2} = \dfrac{1}{4} \Rightarrow x^2 = 6400 \Rightarrow x = 80$ km/h (the positive root in range). Consumption $y = 60 - 20 - 20 = 20$ km per litre. (It is a maximum: for $x < 80$ the gradient is positive, for $x > 80$ negative.)

4 The displacement s metres of a particle from a fixed point O after t seconds is $s = 6\sqrt{t} + 2t + 5$ for $0 \le t \le 4$. Find, when $t = 1$, the particle's **a** velocity in m/s **b** acceleration in m/s². [6]

> **a** Velocity $= \dfrac{ds}{dt}$. Write $s = 6t^{\frac{1}{2}} + 2t + 5$: $v = 3t^{-\frac{1}{2}} + 2 = \dfrac{3}{\sqrt{t}} + 2$. At $t = 1$: $v = 5$ m/s.
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> **b** Acceleration $= \dfrac{dv}{dt} = -\dfrac{3}{2}t^{-\frac{3}{2}} = -\dfrac{3}{2\sqrt{t^3}}$. At $t = 1$: $a = -1.5$ m/s² (slowing down).

Total: 25 marks

Check: multiply by the power, then reduce the power by 1; constants vanish · expand brackets or rewrite $\frac{a}{x^n}$ as ax^{-n} and $\sqrt{x}$ as $x^{\frac{1}{2}}$ before differentiating · stationary point: $\frac{dy}{dx} = 0$; classify by gradient either side · velocity $= \frac{ds}{dt}$, acceleration $= \frac{dv}{dt}$.